STAT 579: Course Schedule

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This page provides a detailed week-by-week schedule for STAT 579. Readings should be completed *before* the first class of the week.

- **HR:** Hernán, M. A., & Robins, J. M. (2024). *Causal Inference: What If*.
- **Neal:** Neal, B. (2020). *Introduction to Causal Inference*.
- **Cunningham:** Cunningham, S. (2021). *Causal Inference: The Mixtape*.
- **VanderWeele:** VanderWeele, T. J. (2015). *Explanation in Causal Inference*.

Part I: Foundations of Causal Inference

Week	Dates	Core Topic	Primary Readings	Optional Reading (Cunningham)	Project Connection
1	Aug 19, 21	What is a Causal Effect?	**HR:** Ch. 1–2; **Neal:** Ch. 1–2	Ch. 2: Potential Outcomes	Define your causal question.
2	Aug 26, 28	DAGs & Assumption Transparency	**Neal:** Ch. 3; **HR:** Ch. 6	Ch. 3: Directed Acyclic Graphs	Draw a preliminary DAG.
3	Sep 2, 4	Exchangeability, Positivity	**HR:** Ch. 7; **Neal:** Ch. 4	Ch. 4: Regression	Identify assumptions for your DAG.
4	Sep 9, 11	Bias (Confounding, Selection)	**HR:** Ch. 8–9	-	Consider biases in your dataset.

Part II: Estimation Methods

Week	Dates	Core Topic	Primary Readings	Optional Reading (Cunningham)	Project Connection
5	Sep 16, 18	Propensity Scores	**HR:** Ch. 12–13; **Neal:** Ch. 7	Ch. 5: Matching & Subclassification	Select adjustment strategy.
6	Sep 23, 25	IPTW & MSMs	**HR:** Ch. 15; **Neal:** Ch. 7.5–7.6	-	Evaluate MSM relevance.

Week	Dates	Core Topic	Primary Readings	Optional Reading (Cunningham)	Project Connection
7	Sep 30, Oct 2	G-computation	**HR:** Ch. 20–21	-	Explore g-methods.
8	Oct 7	Instrumental Variables	**HR:** Ch. 16; **Neal:** Ch. 9	Ch. 8: Instrumental Variables	Assess natural experiments.

Note: *UNM Fall Break is Oct 9–10. Thursday's class is cancelled.*

Part III: Advanced Topics & Applications

Week	Dates	Core Topic	Primary Readings	Project Connection
9	Oct 14, 16	Effect Modification	**HR:** Ch. 4–5; **Neal:** Ch. 10	**Milestone: Project Proposal Due.**
10	Oct 21, 23	Mediation I (Foundations)	**VanderWeele:** Ch. 1–2; **Neal:** Ch. 14	Refine DAG with mediators.
11	Oct 28, 30	Mediation II (Application)	**VanderWeele:** Ch. 3; **HR:** Ch. 23	-
12	Nov 4, 6	Sensitivity Analysis	**Neal:** Ch. 8; **HR:** Ch. 14	Plan robustness checks.
13	Nov 11, 13	Target Trial Emulation	**HR:** Ch. 22	Reframe project as a trial.
14	Nov 18, 20	**Final Presentations**	-	Present your findings.

Note: *Thanksgiving Break is Nov 27-28.*

Finals Week (Dec 8–13): Final Project Paper Due.